Introduction

- Imagine working on your laptop or checking e-mail from anywhere in your home.

- Imagine being able to connect to your office network from an airport or coffee shop.

- Now , imagine doing all these things easily and quickly, without worrying about finding a wired network connection.
  That is ***Wi-Fi*** !



What is Wi-Fi?



- **Wi-Fi** (Wireless Fidelity) is a generic term that refers to the IEEE 802.11 communications standard for Wireless Local Area Networks (WLANs).

- Wireless Technology is an alternative to Wired Technology, which is commonly used, for connecting devices in wireless using radio waves.

- Allows you to access the Internet while on the move ; you can remain online while moving from one area to another, without a disconnection or loss in coverage.

IEEE 802.11 standard



- In 1997, the Institute of Electrical and Electronic Engineers (IEEE) drafted the 802.11 standard for wireless local area networking.

- In 1999, networking hardware companies accepted the standard and began manufacturing products using the 802.11b protocol which operated in the 2.4 GHz range and was capable of transmitting at speeds of 11 Mbps.

- The 802.11a protocol was also released in 1999, operating at 5 GHz with transmissions speeds of 54 Mbps, but its cost was high.

IEEE 802.11 standard (cont.)



Network standard	Maximum Speed (Mbps)	Range (feet)	Frequency (GHz)	Power drain	Cost
802.11b	11	100-150	2.4	Moderate	Low
802.11a	54	60-100	5	High	High
802.11g	54	150-250	2.4	Moderate	Moderate
802.11n	200	Up to 300 feet	2.4 & 5	Moderate	Moderate

Hotspot

What is a Hotspot ?



- A Hotspot is any location where **Wi-Fi** network access is made publicly available.
- Hotspots are equipped with a Broadband Internet connection, and one or more Access Points that allow users to access the Internet wirelessly.
- Hotspots can be setup in any public location that can support an Internet connection. All the locations discussed below are examples of Hotspots.

Wi-Fi Hotspot Locations

- ✓Airports
- ✓Hotels & Resorts
- ✓Restaurants
- ✓Coffee Shops
- ✓Bookstores
- ✓Shopping Malls





- There are several online services allow you find hotspot locations.

Elements of a Wi-Fi Network

- **Access Point (AP)** - The AP is a wireless LAN transceiver or “base station” that can connect one or many wireless devices in the same time to the Internet.
- **Safeguards** - Firewalls and anti-virus software protect networks from uninvited users and keep information secure.
- **Wi-Fi cards (Adapters)** - They accept the wireless signal and relay information. They can be internal and external.

PCI and USB adapters

For Desktop

PCI adapter (Internal)



USB adapter (External)



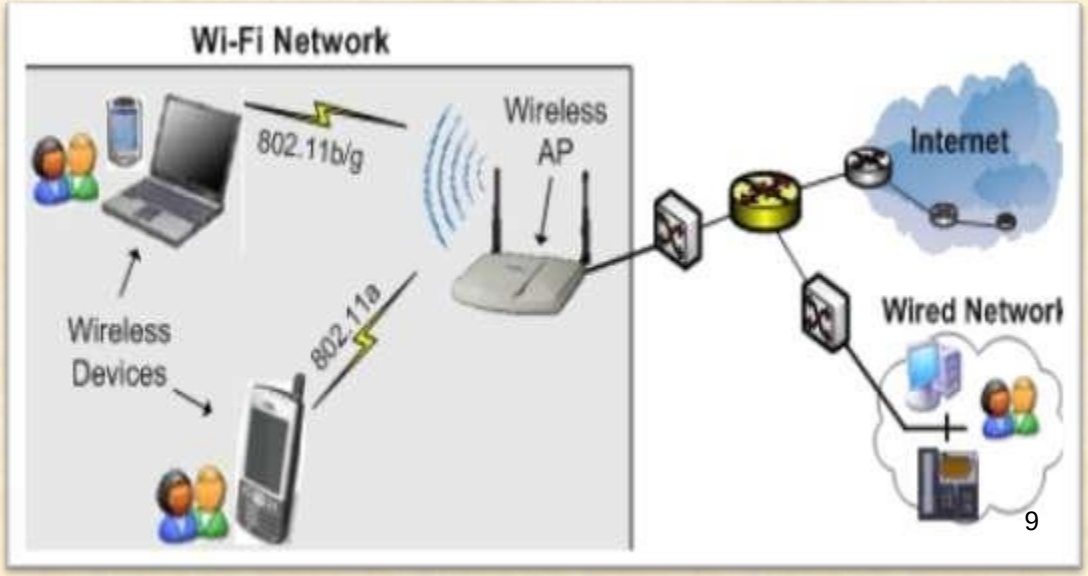
For Laptop

PCMCIA adapter



How a Wi-Fi Network Works ?

- A **Wi-Fi** connection works through a transmitting antenna, which is usually connected to a DSL or cable Internet connection. The antenna on the router will then beam radio signals through a specific range. Another antenna, which is on the laptop or personal computer, receives the signal.



Advantages of Wi-Fi



- **No Wires** - A truly wireless networking solution.
- **No Waiting** - Fast, easy deployments.
- **No Worries** - A wireless networking system that is secure, easy to manage, and built to grow with you.
- **Ease of Installation** - Quick, easy setup.
- **Fast data transfer rates**

Limitations of Wi-Fi



- **Limited range**
- **Data security risks** :a huge challenge for **Wi-Fi** networks.
- **Interference from other devices** : such as telephones, microwave ovens.
- **High power consumption** :making battery life and heat a concern .

Basic Wi-Fi Security Techniques



- **WEP**(Wired Equivalent Privacy) : The original encryption technique specified by the IEEE 802.11 standard.
- **WPA**(Wi-Fi Protected Access): A new standard that provides improved encryption security over WEP.
- **WPA2 :** is an improved version of WPA that uses Advanced Encryption Standard (AES) technology.

Basic Wi-Fi Security Techniques(cont.)



Securing Method	Encryption Type Used	Security Level	Notes
WEP	RC4 encryption algorithm	Low	No longer used; it is can be hacked easily
WPA	TKIP Protocol	High	provides improved encryption security over WEP
WPA2	CCMP Protocol	Very High	An improved version of WPA that uses Advanced Encryption Standard

Conclusion

- **Wi-Fi** is a simple, cost-effective way to connect to the Internet, without the need to physically connecting wires.
- In 1997 IEEE drafted the 802.11 standard for wireless local area networking.
- Hotspot is a geographic area setup in any public location, and has a readily accessible wireless network.
- Security is a huge challenge for **Wi-Fi** Networks, many Security Techniques are used to improve it.
- **Wi-Fi** Networks have a several limitations that should be concerned.

The Future of Wi-Fi

- The future of **Wi-Fi** is very bright. Its growing in popularity because of decreasing costs and the freedom it gives to users.

